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RESEARCH ARTICLE

HyCoViT: Hybrid Convolution Vision Transformer With Dynamic Dropout for Enhanced Medical Chest X-Ray Classification

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ABSTRACT Medical chest X-ray (CXR) classification necessitates balancing detailed local feature extraction with capturing broader, long-range dependencies, especially when working with limited and heterogeneous datasets. In this paper, we propose HyCoViT, a hybrid model that integrates a custom Convolutional Neural Network (CNN) block with Vision Transformers (ViTs). This approach combines the locality of CNN-based latent space representations with the global attention mechanisms of ViTs. To address overfitting in data-scarce scenarios, we introduce a Dynamic Dropout (DD) algorithm that adaptively adjusts the dropout rate during training. Additionally, we enhance model generalization using a combination of traditional data augmentation and MixUp techniques. We evaluate HyCoViT on a multi-class classification task involving COVID-19, pneumonia, lung opacity, and normal CXR images. While COVID-19 serves as a case study, the model's design is generalizable to various medical imaging applications. Experimental results show that HyCoViT achieves state-of-the-art (SOTA) performance, with 98.81% accuracy for three-class surpassing the existing CNN-based model by average +4.90%., and SOTA transformer-based average by 2.05%. In four-class classification, HyCoViT achieves the highest accuracy at 96.56%, which is 8.32% higher than the average accuracy of SOTA CNN-based models and 4.96% higher than the average accuracy of other SOTA transformer-based models. These results surpass many existing CNN-based and transformer-based models, demonstrating the robust generalization capabilities of our method. Furthermore, we provide interpretable, attention-based visualizations that highlight crucial lung regions to support context-aware decisions and ultimately improve patient outcomes.

INDEX TERMS X-ray detection, convolution neural network, vision transformer, dynamic dropout, explainability.

I. INTRODUCTION

Chest X-ray (CXR) imaging serves as a critical diagnostic tool for various pulmonary conditions, ranging from common respiratory infections to more severe diseases that can lead to complications like pneumonia. Identifying such conditions accurately and promptly is essential, as it enables timely interventions, guides treatment strategies, and ultimately improves patient outcomes. However, certain diseases, such as COVID-19, which emerged as a global health crisis, pose additional challenges. COVID-19 shares clinical manifestations like fever, fatigue, and cough with other respiratory ailments [1], making precise and efficient differentiation crucial for preventing severe consequences and mitigating healthcare burdens.

Real-time reverse transcription polymerase chain reaction (RT-PCR) and other laboratory tests are cornerstones in diagnosing COVID-19 and viral infections; however, they are prone to false negatives and often have limitations in detecting early-stage infections [2], [3]. Under such circumstances,

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evaluating CXR images emerges as a complementary or alternative diagnostic route. Conditions like COVID-19 often exhibit telltale pulmonary patterns, such as ground-glass opacities and consolidations, in CXR scans [4], [5]. As a result, CXR analysis can facilitate preliminary screening and triage, guiding physicians toward more targeted testing or treatment plans.

Beyond a single disease, the broader medical landscape faces the challenge of providing reliable interpretations of CXR images, especially in resource-limited settings where the number of trained radiologists may be insufficient. This shortage can delay the diagnostic process and hinder timely care. To address this issue, researchers have increasingly looked to machine learning and Deep Learning (DL) techniques for automated image analysis [6], [7], [8]. These computational approaches aim to reduce the workload on human experts, enhance diagnostic accuracy, and improve healthcare accessibility, particularly in communities where medical imaging resources and expertise are constrained.

In recent years, the emergence of Graphics Processing Units (GPUs) has facilitated the development of DL models. The utilization of deep learning methods, particularly Convolutional Neural Network (CNN), has demonstrated remarkable effectiveness in various computer vision tasks, including image classification, image segmentation, and object detection [9]. Building on this success, significant research efforts have been dedicated to the application of Artificial Intelligence (AI) and DL models in automating image analysis within the clinical domain and more recently chest X-ray classification.

CNN models are advantageous due to the inherent local translation equivariance provided by the convolution operation. This enables them to effectively capture local features that remain invariant to spatial transformations of neighboring pixels [10]. However, this characteristic may result in a bias towards local details, potentially limiting their ability to comprehend the global contextual understanding of the entire image. In contrast, Vision Transformer (ViT) overcomes the limited receptive field constraint of CNNs and excel at capturing long-range dependencies and obtaining a holistic understanding of the entire image [11], [12]. By leveraging self-attention mechanisms to capture global relationships between all patches, ViT-based models offer a more comprehensive perspective at the cost of larger amounts of training data requirement to achieve similar levels of translation equivariance as CNN models. This limits their generalization ability when trained on smaller datasets.

Despite the success of both CNNs and ViT models in developing diagnosis systems, they still face two significant challenges: i) the need for a large amount of training data to avoid overfitting and ensure a well-generalized model, and ii) the inherent ''black box'' nature of these models, which lacks the explainable transparency necessary to assist doctors during medical diagnosis.

The issue of data scarcity is not new in the medical field, particularly with the emergence of new and unknown diseases. The available data are often imbalanced, making it difficult to train DL models effectively and impacting their overall performance. To mitigate these effects, data augmentation is commonly used as a solution. However, if the distribution of the augmented data does not accurately represent real-world data, the resulting images may not always be realistic, leading to a biased model. To address this issue and balance out the potential drawbacks of unrealistic data mixes, this paper proposes combining traditional data augmentation with MixUp data augmentation. MixUp involves creating convex interpolations of input images and their corresponding labels to serve as additional training data. This approach aims to enhance the robustness and generalization of the model. Furthermore, in order to enhance the interpretability of our model, a detailed visualization of the attention weights was generated through an interpretive attention-weight mapping to facilitate an in-depth analysis of the model's focal regions during prediction. This approach allows us to gain valuable insights into the model's decision-making process when classifying COVID-19 CXR, providing a clearer understanding of the regions that most influence the model's predictions. By integrating these methods, we aim to develop a more effective and transparent diagnosis system, capable of aiding medical professionals with reliable and interpretable results.

In summary, our contributions are as follows:

- Introduce a Hybrid Convolution Vision Transformer (HyCoViT) model, which integrates a custom convolutional block with a standard ViT to address the lack of inductive bias in ViT. By placing our lightweight CNN component at the input stage, we effectively refine local features before passing them to the transformer, enabling both localized and global feature extraction.
- Custom Convolution Block is designed for Enhanced Locality and capture Long-Range dependency between image distant regions. This block merges multiple atrous (dilated) convolutions at different dilation rates for multi-scale local feature capture without significantly increasing computation. It also employs group normalization for further stability under small mini-batch sizes, and uses the Mish activation to as a smooth and non-monotonic preserve to preserve gradient flow in deeper networks.
- Mitigate overfitting by integrating classical and MixUp data augmentation techniques, avoiding unrealistic medical images.
- Introduce a novel Dynamic Dropout (DD) algorithm that adaptively adjusts the dropout rate to reduce overfitting.
- Validate our approach on chest X-ray images, providing explainable multiclass classification results that highlight infection locations for improved diagnostic transparency.

Our proposed system accurately identifies the location and extent of COVID-19 infections in lung radiology for both three-class and four-class classification detection scenarios. By integrating explainable classification results, the system offers clear visualizations and insights into the model's decision-making process, enhancing the transparency and reliability of diagnoses for radiologists and other medical experts. This explainability is crucial in building trust and encouraging the adoption of AI-driven diagnostic tools in medical practice.

The remainder of the paper is organized as follows. Section II reviews the related work. In Section III, we first formulate the original vision transformer followed by proposed HyCoViT explanation and break down of its building blocks such as the proposed DD algorithm. Experimental evaluations of our proposed method are provided in Section IV on multi-class chest X-ray detection for COVID-19 compared to state-of-the-art (SOTA) methods. In section V, we discussed the current limitations and potential improvement for future work. We conclude this paper in Section VI.

II. RELATED WORKS

Deep learning techniques, particularly CNNs and ViTs, have shown significant promise in medical imaging applications. These models are widely used to address the challenges of extracting both local features and long-range dependencies in diagnostic tasks, especially in scenarios with limited and heterogeneous datasets.

A. CNN-BASED APPROACHES

CNNs are renowned for their ability to capture detailed local features in medical images. Various CNN architectures, such as VGG [13], DenseNet [14], and ResNet [15], have been employed for tasks ranging from pneumonia detection to lung cancer classification. For example, a CNN-based model developed by Rajaraman et al. [16] iteratively pruned CNN layers to achieve optimized performance on CXR classification tasks. Similarly, the CNN-ConvLSTM model combined a ResNet152 backbone (pre-trained on ImageNet) with a four-layer ConvLSTM architecture to capture both spatial and temporal features, achieving high accuracy in medical image classification [17]. Due to the urgency of recent global health challenges, the current application of these methods has focused on specific respiratory infections; however, this framework can be effectively extended to address other diagnostic challenges, such as tuberculosis and pneumonia detection [18], [19].

B. ViT-BASED APPROACHES

ViTs have recently gained prominence for their ability to capture long-range dependencies through self-attention mechanisms [20]. In medical imaging, ViTs have been adapted to detect subtle variations in texture and structure, which are critical for accurate diagnosis. Sitaula and Hossain [21] introduced an attention-enhanced ViT model integrated with VGG-16, achieving promising results in multi-class classification tasks. Gao et al. [22] applied a ViT model to COVID-19 detection and highlighted its ability to capture global contextual features. Furthermore, PneuNet [23] utilized multi-head attention applied to channel patches rather than feature patches, allowing the model to discern subtle lung texture variations effectively. These models illustrate the versatility of ViTs in capturing complex patterns, making them suitable for diverse medical imaging applications, including breast cancer detection [24] and brain tumor classification [25].

C. HYBRID CNN-ViT APPROACHES

Hybrid models that combine CNNs and ViTs leverage the strengths of both architectures, capturing both local and global features. Chen et al. [17] introduced the CMT (CNN-Transformer) framework, which cascades CNNs (e.g., ResNet, ConvNext) with transformers, achieving accuracies between 93.42% and 95.28% in medical image classification tasks. Similarly, xViTCOS-CXR [26] combined CNNs with ViTs using a multi-stage transfer learning approach to achieve 96% accuracy in the four-class classification of COVID-19, pneumonia, lung opacity, and normal CXR images. This hybrid approach successfully captures both fine-grained local details and long-range dependencies, enhancing performance in scenarios where dataset sizes are limited or heterogeneous [27], [28].

These studies demonstrate that hybrid models can be effectively generalized to various medical imaging tasks, including retinal disease detection [29] and histopathological image analysis [30].

D. APPLICATION OF HYBRID MODELS IN REAL-WORLD MEDICAL IMAGING

The success of hybrid models in COVID-19 detection illustrates their potential for broader medical imaging applications. For example, the methods developed for COVID-19 classification have been successfully adapted for tuberculosis detection [31], Pneumonia screening [32], and even breast cancer diagnosis [24]. Additionally, as more recent application, DL-based segmentation methods have shown strong performance in liver and abdominal imaging. Dense-PSP-UNet has been proposed for real-time liver ultrasound segmentation, addressing noise and pre-processing challenges [88]. Likewise, Res-PAC-UNet offers efficient liver CT segmentation using pyramid atrous convolutions and lightweight architecture [89]. A recent survey highlights the effectiveness of U-Net variants with specialized modules in handling low contrast and anatomical variability in ultrasound images [90]. These applications highlight the robustness of combining CNN-based local feature extraction with ViT-based global attention, particularly in data-scarce environments where model generalization is crucial. The proposed HyCoViT model extends these advancements by integrating a custom CNN block and ViT architecture

alongside a Dynamic Dropout (DD) algorithm and advanced data augmentation techniques. This design enables HyCoViT to achieve state-of-the-art performance in multi-class classification tasks, making it suitable for a wide range of medical imaging applications beyond COVID-19 detection. A comparative summary of the most relevant state-of-the-art methods is presented in Tables 4 and 5 in the Results section, which provide side-by-side evaluations of classification performance across multiple metrics for 3-class and 4-class classification tasks on X-ray datasets.

III. METHOD

A. ORIGINAL VIT ARCHITECTURE

The ViT, originally derived from its application within NLP models such as BERT [33], has emerged as a powerful tool for image analysis [20]. The success of ViT is widely attributed to the self-attention (SA) mechanism [34], which efficiently captures long-range dependencies by determining the relative importance of each token (or patch embedding) with respect to all other tokens in the sequence.

ViT processes input images x, characterized by height (H), width (W), and the number of channels (C), where $x \in \mathbb{R}^{H\times W\times C}$. The image is split into a sequence of non-overlapping, flattened two-dimensional patches of size $P\times P$. This operation yields $N = \frac{H\times W}{P^2}$ patches, transforming the image data into a format analogous to the structure used in NLP. These patches are then linearly projected into a D-dimensional embedding space using a trainable linear projection layer, resulting in a matrix representation $X \in \mathbb{R}^{N\times D}$.

The SA mechanism operates by capturing interactions among these N embeddings using three learnable weight matrices, which transform the input X into Queries ($W_Q \in \mathbb{R}^{D\times D_q}$), Keys ($W_K \in \mathbb{R}^{D\times D_k}$), and Values ($W_V \in \mathbb{R}^{D\times D_v}$), where $D_q = D_k$. The input sequence X is projected onto these weight matrices, obtaining $Q = XW_Q$, $K = XW_K$, and $V = XW_V$. The resulting attention matrix $A \in \mathbb{R}^{N\times N}$ is computed as follows [34]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{D_k}}\right)V \tag{1}$$

Multi-Head Self-Attention (MHSA) extends the SA mechanism by allowing the model to focus on different parts of the input sequence simultaneously, thus capturing a richer set of dependencies within the data. By using multiple attention heads, the model learns to attend to information from various subspaces at different positions, leading to improved performance and generalization. MHSA is composed of multiple attention heads, each with its own set of learnable weight matrices W_i^Q, W_i^K, W_i^V, where $i = 0, \ldots, h-1$, and h denotes the number of heads. The MHSA process can be formalized as:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{2}$$

where each individual attention head is computed as:

$$\text{head}i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{3}$$

The outputs from the heads are concatenated and projected via a learnable weight matrix W^O, resulting in:

$$\text{MHSA}(Q, K, V) = \text{Concat}(Z_0, Z_1, \ldots, Z_{h-1})W^O \tag{4}$$

A trainable linear projection matrix E is used to map these reshaped patches into a D-dimensional space, generating a sequence of embedded image patches. A learnable class token X_{class} is prepended to the sequence of embedded patches. This class token is inherently associated with the output classification, Y. In the final step, positional embeddings E_{pos}, which provide spatial information, are added to the patch and class embeddings.

The resulting sequence of vectors, denoted by z_0, serves as input to the transformer encoder. Let $X \in \mathbb{R}^{n\times d}$ represent the token features, where n is the sequence length and d is the embedding dimension:

$$z_0 = [X_{\text{class}} + E^0_{\text{pos}}; x_p^1 E + E^1_{\text{pos}}; x_p^2 E + E^2_{\text{pos}}; \ldots; x_p^N E + E^N_{\text{pos}}] \tag{5}$$

In this approach, we utilize a transformer encoder network consisting of L identical layers. Each layer includes a MHSA module and a feedforward network. We employ $h = 12$ parallel attention heads. The final classification is produced by the class token from the L-th encoder layer. The classification head consists of a Multilayer Perceptron (MLP) with a single hidden layer, using the Gaussian Error Linear Unit (GELU) activation function. Layernorm (LN) is applied before every block, and residual or skip connections after every block. The intermediate computations at each layer are as follows:

$$z'_l = \text{MHSA}(\text{LN}(z_{l-1})) + z_{l-1}, \quad l = 1, \ldots, L \tag{6}$$

$$z_l = \text{MLP}(\text{LN}(z'_l)) + z'_l, \quad l = 1, \ldots, L \tag{7}$$

$$y = \text{LN}(z_L^0) \tag{8}$$

The original ViT architecture draws heavily from the transformer model introduced for NLP tasks [33], with adaptations to handle image data. ViT processes image patches of 16×16 pixels, augmented with positional embeddings and a class token. There is a trade-off between patch size, performance, and computational cost. Smaller patches generally improve accuracy but increase computational overhead. In line with [20], a patch size of 16×16 was chosen for this work to balance computational efficiency and performance. The learnable class token is passed through the classification head to produce the final prediction based on its state. The overall architecture of ViT, as part of our proposed HyCoViT model, is depicted in Figure 1.

B. PROPOSED HYCOVIT ARCHITECTURE

Despite the success of both CNN and ViT models at large scales, their performance often degrades in data-scarce medical imaging scenarios. Vision Transformers (ViTs) leverage

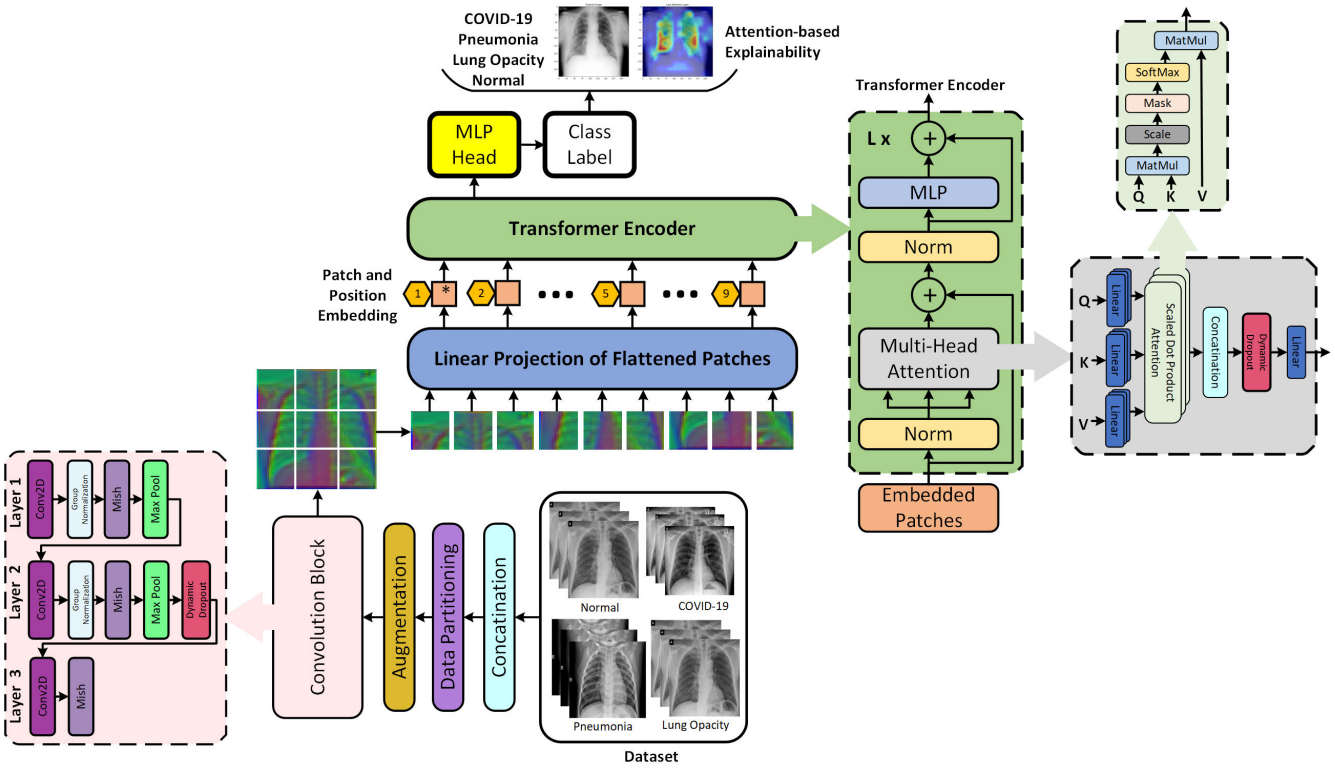


FIGURE 1. Proposed HyCoViT architecture[1].

self-attention to capture long-range dependencies, enabling them to focus on relevant image regions during classification. However, they inherently lack the inductive biases (e.g., locality and translation invariance) that CNNs naturally possess, an absence that can hinder the extraction of fine-grained local features when training data are limited. In contrast, CNNs embed strong priors about local receptive fields, shared weights, and spatial subsampling, thus excelling at detailed feature extraction. Yet, CNNs can overlook broader context or distant pixel relationships, which ViTs inherently model through self-attention. To address these challenges, we propose HyCoViT, a hybrid architecture that integrates a custom convolution block with the ViT encoder. This design injects local inductive biases at an early stage, combining CNN-style feature extraction with ViT's global context modeling. The custom convolution block, detailed in Table 1, employs multiple dilated convolutions (dilation rates of 1, 2, and 4) to expand the receptive field without incurring substantial computational overhead to provide multi-sclare feature extraction capabilities. This multi-dilation strategy enables the detection of subtle lesions (e.g., minor opacities), while group normalization ensures stable training under small or imbalanced mini-batches commonly found in medical datasets. Furthermore, the Mish activation preserves slight negative values, enhancing gradient flow and aiding the capture of low-intensity features in chest X-ray images. After passing through this convolution block, low-level feature maps are patchified and fed into the ViT encoder for global self-attention. By fusing the locality of CNNs with the long-range representational power of ViTs, HyCoViT provides an end-to-end solution for multi-class chest X-ray classification. This hybrid approach effectively overcomes the data-hungry nature of standard ViTs, offering robust performance in limited-data settings while retaining essential fine-grained and contextual information for accurate disease detection.

In this model, we use a ViT with transformer encoder which includes 12 layers of self-attention and feed-forward networks, followed by final layer normalization and classification head with output legits for multi-class classification task. It has been reported [35] that simply increasing depths basically introduces more parameters and this leads to cumbersome computational cost and more importantly overfitting problem in small datasets. To mitigate the computational cost, we employ the Mish activation function in lieu of ReLU as a frequent choice in DL applications. See Section III-C for more in-depth explanation of Mish. Further, we apply Group Normalization (GN) in the proposed convolution block to normalize the activations within each group of channels instead of normalizing across the entire batch or channel dimension. This allows the network to capture group-specific statistics and reduces the dependency on the batch size [36].

[1]The code is available at: https://github.com/DIMOMID/HyCoViT.git

Additionally, In this paper, we propose Dynamic Dropout (DD) algorithm as a simple and effective method to control the model complexity by adaptively adjust the dropout rate during each epoch inside our proposed convolution block as well as on ViT attention layer - see Figure 1 for DD block shown in red. Section III-D explains DD algorithm details. Table 1 presents the architecture of the proposed convolution block as part of the HyCoViT.

To determine the optimal placement of DD within the custom CNN block, we experimented with inserting it after each of the three convolutional layers. As shown in Ablation Study Table 8, placing DD after the second convolutional layer (L2) provided the best performance. This position offers a balance between preserving low-level structural features (captured by the first layer) and effectively regularizing more abstract, task-specific representations learned in deeper layers. Empirically, this configuration led to more stable convergence and higher accuracy compared to applying DD after either the first or third layer.

Due to lack of sufficient relevant dataset for training in specific tasks we employ the transfer learning as a common practice to avoid training a large-scale network from scratch [37]. This approach is a two-step process. At first, a model is trained on large volumes of benchmark datasets such as ImageNet [38]. Then we fine-tune the model based on the specific task at hand, which in our case is multi-classification task in CXR images. Transfer learning applies knowledge from a large pre-trained dataset to new tasks with limited data by fine-tuning on smaller, specialized datasets. This approach requires a fraction of the data required to train a model from scratch and can be trained in a shorter duration, with less computational resources. Transfer learning tactic allows the HyCoViT model to perform well on both smaller and larger datasets.

To address the imbalanced nature of the medical CXR dataset, we utiliz the focal loss function as our loss function [39]. Class weights w_t are computed based on the distribution of class labels to emphasize harder-to-classify examples. The focal loss function is defined as follows:

$$\mathrm{FL}(p_t) = -w_t(1-p_t)^{\gamma}\log(p_t) \tag{9}$$

where p_t is the model's estimated probability for the true class, w_t is a weighting factor for the class t, and γ is a focusing parameter that adjusts the rate at which easy examples are down-weighted. As opposed to Weighted Cross Entropy, which assigns greater weights to underrepresented classes, Focal Loss reduces the loss contribution for well-classified and easy examples by emphasizing hard-to-classify ones through the incorporation of a modulating factor $(1-p_t)^{\gamma}$ in conjunction with class weights (w_t). In this study, we set γ to 2 to effectively balance the focus on difficult samples without causing training instability, thereby enhancing model performance in imbalanced classification tasks.

TABLE 1. Convolution Block architecture. L: Layer, K: Kernel size, S: Stride, P: Padding, D: Dilation, GN: Group Normalization, M: Mish, MP: Max Pooling, DD: Dynamic Dropout.

L1	Con2D(3, 32, K3, S1, P1, D1)	GN	M	MP(K2,S2)	-
L2	Con2D(32, 64, K3, S1, P1, D2)	GN	M	MP(K2,S2)	DD
L3	Con2D(64, 3, K3, S1, P1, D4)	-	M	-	-

C. MISH ACTIVATION FUNCTION

The choice of activation function (AF) has a significant effect on the performance of training dynamics in deep networks. The Rectified Linear Unit (ReLU) function is a popular choice as an AF in many neural network applications [40], [41]. However, while the ReLU function is simple and effective, it suffers from issues such as under-utilization of negative values, resulting in a vanishing gradient as its output becomes zero for all negative values (known as dying ReLU). Additionally, it has limited non-linearity, exhibiting a singularity at zero, and unbounded output, which may lead to training instability [42], [43]. As a result, its application in deeper networks can be hampered, and the learning process can be inhibited.

To address these drawbacks, Mish was introduced by [44] as a smooth, continuous AF without zero discontinuities. This improves gradient propagation during optimization. Additionally, as a non-monotonic activation function, Mish helps preserve small negative weights in deep neural network architectures. Being bounded below and above with a range of $[-0.31, \infty)$ is advantageous since it results in strong regularization effects. Mish AF is mathematically defined as [44]:

$$f(x) = x\tanh(\mathrm{softplus}(x)) = x\tanh(\ln(1+e^x)) \tag{10}$$

and its first derivative is calculated as below:

$$f'(x) = \frac{(e^x\omega)}{\delta^2} \tag{11}$$

where $\omega = 4(x+1) + 4e^{2x} + e^{3x} + e^x(4x+6)$ and $\delta = 2e^x + e^{2x} + 2$.

Mish tends to produce a smoother loss landscape, which can lead to convergence to wider minima, improving generalization compared to ReLU. ReLU's tendency to produce sharp transitions and its zero-gradient issue for negative values may lead to narrower local minima [44]. "Wider minima" refer to regions in the loss landscape where the loss remains low across a broader range of parameter values, leading to better generalization as the model's performance is less sensitive to small parameter changes. Figure 2 represents a comparison of Mish and ReLU and their first derivatives. In this study, the Mish AF is used.

D. DYNAMIC DROPOUT

Deep Neural Network (DNN) architectures have increased in depth and size in pursuit of superior performance and model capacity gain which often led to comprising hundreds of layers and millions of parameters. However, this complexity

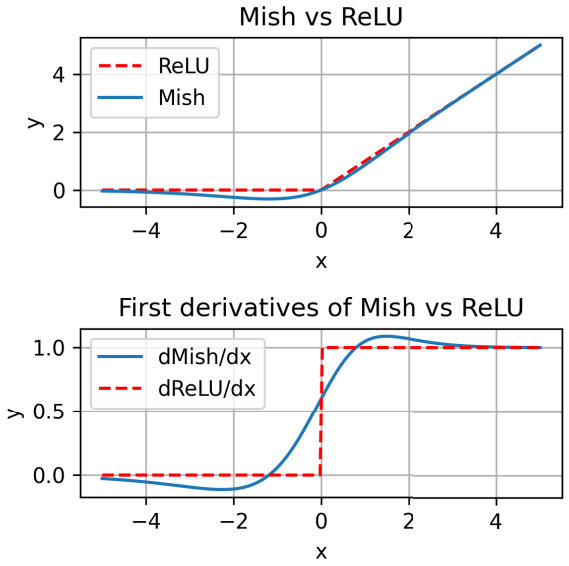


FIGURE 2. Comparison of Mish and ReLU and their first derivatives.

is susceptible to overfitting, and therefore, regularization techniques such as early stopping [45], L1 and L2 weight penalties, batch normalization, and dropout [46], [47] are essential for the training process of DNN models. Dropout is a technique originally introduced by Srivastava et al. [48], where each neuron is deactivated with a fixed probability p according to a Bernoulli distribution $m_i \sim \text{Bernoulli}(p)$. This method helps address the overfitting issue during the training of neural networks by preventing co-adaptation of neurons. More specifically, during each training iteration, based on a fixed probability, a set of neurons are sampled and deactivated, then the training is performed on the resultant subset of neurons. As a result, the network can be thought of as a collection of many different subsets of neurons that share parameters. This ensemble of subsets enhances the overall performance of the network. Additionally, dropout regularizes the network by preventing neurons from co-adaptation with each other, therefore, each neuron individually provides information without relying on the synergy or collaboration of other neurons, which leads to improved generalization. Despite the effectiveness of the standard fixed dropout ability in regularizing many DNN models, still there is an opportunity to improve it for better convergence and overall performance. Motivated by standard dropout method [48] and recently introduced stochastic dropout method [47], where the dropout rate stochastically varies in each iteration, we propose algorithm DD in this study. The drawback of the fixed probability dropout rate as a hyperparameter is that it requires several trial-and-error to set a conservative value as a suboptimal parameter which is an expensive choice in large DNN for which training may last as long as several days. On the other hand, stochastic dropout rate presented some promising results; however, a large jump and sudden change in the dropout rate in every iteration means the model is forced to suddenly forgetting a large proportion of its learned features, which can cause instability in the learning process. The model may struggle to adjust to these drastic changes, and this could result in unstable training loss, volatile validation metrics, and, in the worst case, the model may fail to converge or not be as effective as anticipated as a regularization technique.

The DD algorithm incorporates an adaptive mechanism to regulate the dropout rate during training by leveraging a feedback loop between the Validation-Weighted Generalization Gap ($VWGG$) and its Percentage Success Rate (PSR). The $VWGG$ metric serves as a critical measure of the trade-off between validation loss and the model's generalization capability, capturing both absolute and relative trends in performance. It is defined as:

$$VWGG(i) = \lambda \cdot RSVLA(i) + (1 - \lambda) \cdot RGGC(i) \tag{12}$$

where $\lambda \in [0, 1]$ is a weighting parameter (set to 0.7) that balances the contributions of the Recent Scaled Validation Loss Average ($RSVLA$) and the Relative Generalization Gap Change ($RGGC$). The $RSVLA$ quantifies the normalized recent validation loss averaged over a window of size (w) spanning 3 epochs. It is scaled relative to the initial validation loss recorded in the first epoch, ensuring comparability across training iterations. The $RSVLA$ is computed as:

$$RSVLA(i, w) = \frac{\frac{1}{w}\sum_{j=0}^{w-1} valid_{loss}(i-j)}{valid_{loss}(i=0)} \tag{13}$$

The $RGGC(i)$, on the other hand, quantifies relative fluctuations in the model's generalization error over successive epochs:

$$RGGC(i) = \frac{MovAvgGE(i) - MovAvgGE(i-1)}{MovAvgGE(i-1)} \tag{14}$$

where $MovAvgGE$ represents the moving average of the Generalization Error (GE) computed over a window size of w:

$$MovAvgGE(GE, i, w) = \frac{1}{w}\sum_{j=1}^{w-1} GE(i-j) \tag{15}$$

and GE represents the root mean square error ($RMSE$) between the training and validation losses. It serves as an indicator of the model's overfitting or underfitting conditions, effectively capturing discrepancies in performance between training and validation data.

The $PSR(\%)$ measures the percentage change in $VWGG$ across consecutive epochs, serving as a normalized indicator of training progress:

$$PSR(i)(\%) = \frac{VWGG(i) - VWGG(i-1)}{VWGG(i-1)} \times 100. \tag{16}$$

The calculation of $PSR(i)$ ensures scale invariance by expressing changes in $VWGG$ as relative percentages, which

Algorithm 1 Dynamic Dropout (DD) Algorithm

Require: $w = 3,\ BS = 0.2,\ PSRT = 5$
$DR_{\min} = 0.01,\ DR_{\max} = 0.99$
DR_i = dropout rate for *i-th* epoch

```
Initialize DR_{i=1} ← 0.5, MovAvg ← [ ]
for i ← 1 to Max_Epoch do
    Train model with DR_i
    Compute VWGG_i per equation (12)
    Compute PSR_i per equation (16)
    Compute SF per equation (17)
    if |PSR_i(%)| < PSRT then
        DR_{i+1} ← DR_i
    else if PSR_i(%) ≥ PSRT then
        DR_{i+1} ← DR_i + BS × SF
    else if PSR_i(%) < −PSRT then
        DR_{i+1} ← DR_i − BS × SF
    end if
    DR_{i+1} ← min(max(DR_i, DR_min), DR_max)
    Update DR_{i+1} in the HyCoViT layers
end for
```

facilitates the application of consistent thresholds across diverse datasets and model architectures.

The DD algorithm adjusts the dropout rate (DR_i) at each epoch using a fixed base-step (BS) for consistent updates, scaled by a dynamic step-factor (SF) that reflects the model's generalization performance. The (SF), is calculated based on scaled sigmoid function as:

$$SF(VWGG, \alpha) = \frac{1}{1 + \exp(-\alpha \cdot VWGG)}, \tag{17}$$

where $\alpha = 1$ is a scaling parameter that controls the steepness of the sigmoid curve. This scaled sigmoid function ensures smooth, bounded adjustments to the dropout rate, avoiding abrupt changes while maintaining responsiveness.

The updated dropout rate is computed as:

$$DR_{i+1} = DR_i \pm SF \times BS, \tag{18}$$

The dropout rate adjustments is determined by the magnitude and direction of $PSR(\%)$. Specifically:

- If $|PSR(\%)| < PSRT$, where $PSRT$ is a predefined threshold (e.g., 5%), the dropout rate remains unchanged to prevent unnecessary fluctuations.
- If $PSR(\%) > PSRT$, the dropout rate increases to enhance regularization and mitigate overfitting.
- If $PSR(\%) < -PSRT$, the dropout rate decreases, allowing the model to better fit the training data and avoid underfitting.

By dynamically balancing regularization strength based on the model's performance, the DD algorithm provides a robust and adaptive mechanism to optimize generalization. Its integration of $VWGG$ and PSR enables it to respond to meaningful trends in training dynamics, enhancing stability and performance across a wide range of applications.

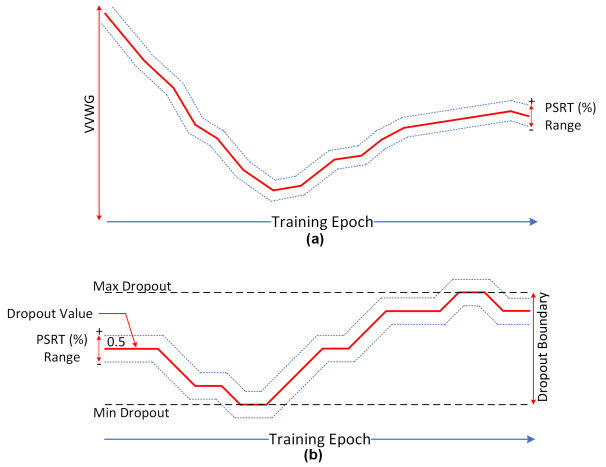


FIGURE 3. Visualization of the proposed dropout algorithm.(a) VVWG changes over training epochs within the PSRT (%) range.(b) Dynamic dropout value adjustments within the PSRT (%) range, bounded by the minimum and maximum dropout values during training epochs.

Algorithm 1 presents the proposed DD algorithm. Figure 3 illustrates the logic of Algorithm 1. The figure shows the PSRT(%) range (dashed lines) relative to $VWGG$. Within this range, the dropout rate remains unchanged; outside this range, the dropout rate is updated. The DD provides a flexible and data-dependent way to regulate the dropout ratio during training process, making it a promising strategy to enhance model performance in complex DL architectures such as proposed HyCoViT. We conducted an abaltation study in section IV-F to assess the effectiveness of the proposed DD in HyCoViT architecture.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

In this section, we first overview the dataset, data augmentation, evaluation metrics, and experiment setup, then we proceed with discussion on the results.

A. DATASET

To ensure a comprehensive and equitable evaluation, we apply the proposed method to a contemporary and clinically significant medical application: COVID-19 detection using chest X-ray images. The dataset utilized in this research is referred to as the COVID-19 Radiography Dataset [49] which includes medical chest radiographs, classified into four distinct categories: Normal, Pneumonia, Lung Opacity (Non-COVID lung infection), and COVID-19. Figure 4 presents a CXR sample for each class. These images were sourced by researchers affiliated with Qatar University in Doha, Qatar, and University of Dhaka, Bangladesh, along with medical practitioners and researchers from Pakistan and Malaysia. The categorization comprises of 3,616 COVID-19 instances, 10,192 Normal instances, 6,012 Lung Opacity instances, and 1,345 Pneumonia instances [49]. The images are stored in the PNG (Portable Network Graphics) format, boasting a resolution of 299 × 299 pixels. Data augmentation is applied

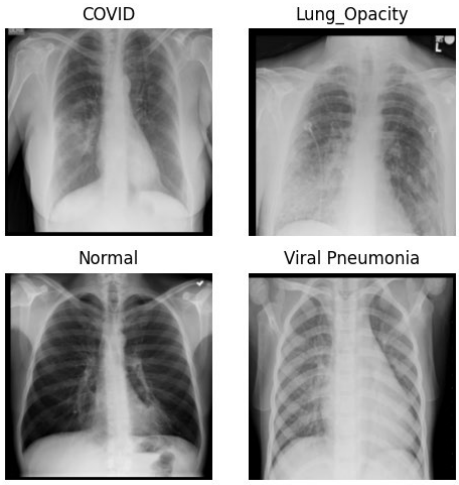


FIGURE 4. Example CXR images for the dataset classes. COVID-19, Normal, Lung Opacity (Non-COVID infection), and Viral Pneumonia.

to the dataset to synthesize sufficient amount of the data for large size models like ViT to prevent overfitting.

B. DATA AUGMENTATION

Despite great breakthrough of deep learning methods such as CNN and ViT in image classification, the performances of these large size networks with millions of parameters rely on sufficient amount of the training data to avoid overfitting. Data augmentation is a data-space solution to the problem of data scarcity and overfitting phenomenon [50], [51].

This strategy increases the diversity of limited available data for training models to generalize better to unseen data, without actually collecting new data by only applying traditional transformation to the existing data. These transformations can include rotations, translations, flips, changes in brightness for images, etc. However, the specific augmentation should be relevant to the nature of task. In X-ray imaging context, unrealistic transformations (e.g., color changes or horizontal flips) could alter important medical information in the image and confuse the model rather than being meaningful transformation [51], [52]. More specifically, for instance, the transformations ''horizontal flip'' and ''vertical flip'' involve mirroring the image along its horizontal or vertical axis, respectively. While these transformations could be useful when working with natural images (e.g., pictures of cats, cars, etc.), they could lead to anatomically incorrect images when applied to X-ray images. The reason is that the human body is not symmetric from left to right or from top to bottom. Therefore, an X-ray image of the lungs, for example, would look different if it was flipped horizontally or vertically, and this could confuse a model trained to detect COVID-19. In other words, when applying data augmentation techniques to medical images, it's important to respect the inherent structure and properties of the data [52], [53].

Despite the benefits of traditional data augmentation techniques, which typically apply transformations to existing individual images without altering the underlying class relationships, deep models can sometimes overfit to the augmented data because the augmented samples are still closely related to the original data, thus not providing enough new information for the model to generalize better. To overcome this issue, MixUp has been introduced by [54]. In MixUp, convex interpolations of training input and their labels are used instead of the original data for training, with the effect of improved generalization and robustness against adversarial attacks. The following describes the MixUp process:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j \tag{19}$$

$$\tilde{y} = \lambda y_i + (1 - \lambda)y_j \tag{20}$$

where x_i, x_j are the input vectors, y_i, y_j are their one-hot coded labels, and $\lambda \in [0, 1]$ is sampled from the Beta distribution: $\lambda \sim \beta(\alpha, \alpha)$ during every mini-batch, where α is a hyperparameter. Lower α leads to more even proportions between λ and $1 - \lambda$.

MixUp creates entirely new data points by mixing images and their labels, significantly increasing the diversity of the training data. This process not only enhances the robustness and generalization of the model but also reduces the risk of the model memorizing specific features of individual training samples. By generating a broader range of training examples, MixUp ensures that the model learns more generalized features. However, the interpolating between pairs in MixUp can sometimes result in unrealistic images or ambiguity in labeling that represent a combination of the four classes.

To address these issues, we balance out the MixUp with traditional data augmentation. In this study, for training set, we first apply traditional augmentation, then apply MixUp to only 20% of training dataset to avoid creating very difficult and unrealistic meaningless training dataset. This sequence ensures that the base images are realistic and diverse before they are mixed. We apply random crop after resizing to a square image of size 224×224, random 2D affine transformation to shift the image horizontally, vertically and also normalization. However, for validation and testing sets, the transformations are appropriately simpler with only resizing and normalizing images so that these datasets represent the ''real-world'' data that the model will encounter and therefore shouldn't be augmented in the same way as the training data.

C. EVALUATION METRICS

In this study, we use conventional assessment metrics typically employed in the literature for classification task [55]. The study presents results through the metrics such as accuracy, precision, recall, and F1-score. Each is defined using true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Accuracy, as shown in Equation 21, quantifies the correctly classified samples (TP

TABLE 2. Three-class classification results per-class (%).

Metric	COVID-19	Normal	Pneumonia	Overall
Accuracy	99.30	98.95	99.38	98.81
Precision	97.89	99.60	95.50	98.78
Recall	99.29	98.82	97.45	98.90
F1 Score	98.58	99.21	96.46	98.84

and TN) in relation to the total sample count.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{21}$$

Precision and recall, commonly used in detection problems, provide a complementary perspective to accuracy. Precision, defined in Equation 22 quantifies the accuracy of positive predictions by dividing the count of correctly classified positive samples (TP) by the total predicted positives (TP and FP).

$$Precision = \frac{TP}{(TP + FP)} \tag{22}$$

Conversely, recall, or sensitivity, presented in Equation 23, computes the ratio of correctly classified positives to all actual positives (TP and FN), indicating the model's propensity to identify infected cases [55].

$$Recall = \frac{TP}{(TP + FN)} \tag{23}$$

Lastly, the F-measure or F1 score is typically considered a balanced measure between precision and recall. It is derived from the weighted harmonic mean of both these parameters, as detailed in Equation 24.

$$F1 - score = \frac{2 \times (Precision \times Recall))}{(Precision + Recall)} \tag{24}$$

D. EXPERIMENTAL SETUP

In our study, we utilized Python 3.6, PyTorch 1.10, and the NVIDIA Tesla V100-SXM2-16GB GPU for model implementation and training. The model's training was performed using the AdamW optimizer accompanied by a step decay scheduler [56]. The initial configurations for the learning rate, batch size, and weight decay were established as 1e-4, 32, and 1e-6, respectively. Given the inherent imbalance typically present in medical image datasets, we employed focal loss as the loss function, whereby distinct class weights were calculate based on the class label distribution [39]. We utilized the pre-trained "vit-base-patch16-224" model [20], originally trained on ImageNet, as the foundation for the ViT component in our HyCoViT model. An early stopping strategy, with a patience of 10, was employed to monitor the training process to enhance optimal model performance. The dataset was partitioned with a ratio of 70%, 15%, and 15% for the training, validation, and testing subsets, respectively. Additionally, to enhance the model's generalization capability and robustness, we applied an appropriate augmentation strategy to the training set as described in Section IV-B.

E. RESULTS AND DISCUSSIONS

1) ANALYSIS STRUCTURE

To evaluate the performance of HyCoViT in detecting COVID-19 from chest X-ray images, we conducted experiments across two multi-classification scenarios, comparing it against other deep learning-based methods. In the first scenario, we developed a classification model to differentiate between three classes: normal, viral pneumonia, and COVID-19. This scenario reflects the real-world challenge of accurately distinguishing COVID-19 from pneumonia, particularly during flu season when the symptoms of pneumonia and COVID-19 are notably similar, complicating the classification process. In the second scenario, we trained HyCoViT to handle a more complex four-class classification task, distinguishing between normal, viral pneumonia, lung opacity, and COVID-19. The subtle overlap of symptoms among the three non-normal classes exacerbates the challenge of accurately detecting COVID-19, emphasizing the complexity of this diagnostic task.

2) THREE-CLASS CLASSIFICATION RESULTS PER-CLASS

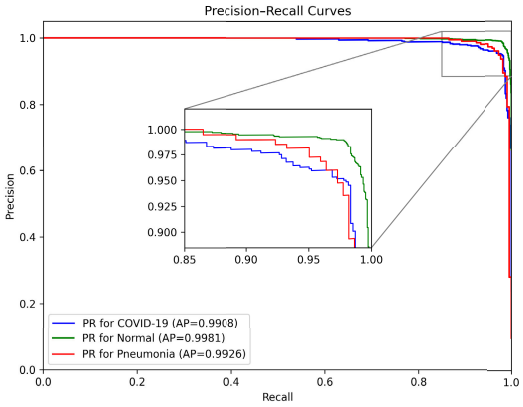
Table 2 presents the per-class results based on evaluation metrics for HyCoViT in the three-class COVID-19 detection scenario, while Figure 6(a) illustrates the corresponding confusion matrix. As observed, HyCoViT exhibited exceptional performance in this scenario, particularly in identifying COVID-19 cases, where it accurately classified 556 instances with minimal misclassifications (only 12 instances in total). The model also demonstrated high accuracy in identifying 1501 normal cases and 191 viral pneumonia cases, further emphasizing its robust overall accuracy. Misclassifications were infrequent, with the most significant being the 11 normal cases incorrectly classified as COVID-19. Overall, the HyCoViT algorithm excels in accurately detecting COVID-19 and effectively distinguishing between different medical conditions. According to Table 2, the overall accuracy of the model stands at 98.81%, with class-specific accuracy of COVID-19 at 99.30%. The model's overall precision, recall, and F1 score are 98.78%, 98.90%, and 98.84%, respectively. These results confirm the HyCoViT's ability to correctly detect positive COVID-19 cases while minimizing false positives in the three-class detection task.

Figure 7(a) visualizes the ROC results reported in Table 2 for HyCoViT, where a one-vs-all strategy was employed to construct individual ROC curves for each class. The resulting ROC curves achieved perfect AUC scores of 1.00 for all three classes, underscoring the model's exceptional discriminative performance.

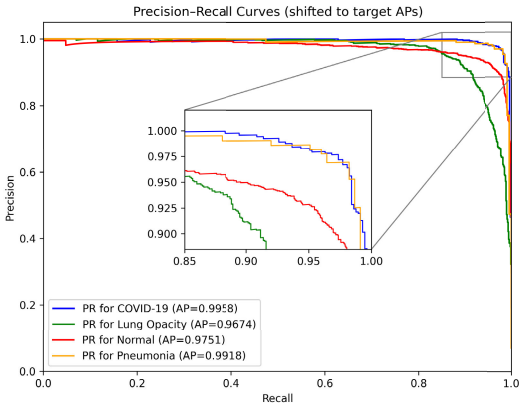
To further strengthen the evaluation of HyCoViT under real-world clinical conditions, we also incorporated Precision–Recall (PR) curves alongside the ROC curves to assess classification performance in both 3-class and 4-class scenarios. Given the class imbalance inherent in chest X-ray datasets, particularly in categories such as

TABLE 3. Four-class classification results per-class (%).

Metric	COVID-19	Lung Opacity	Normal	Pneumonia	Overall
Accuracy	98.53	95.53	96.78	97.40	96.56
Precision	97.89	98.10	98.46	97.40	97.96
Recall	98.58	97.33	98.26	100	98.54
F1 Score	98.23	97.71	98.36	98.68	98.24

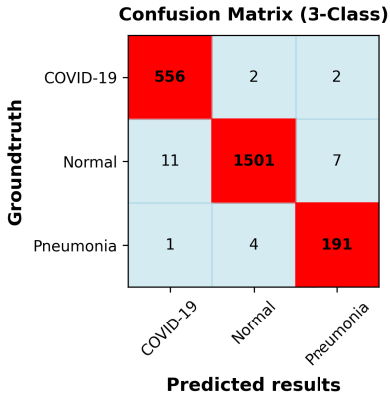


(a) Three-class scenario

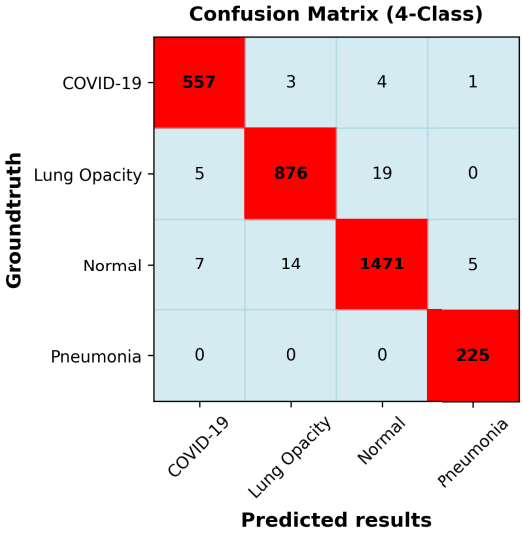


(b) Four-class scenario

FIGURE 5. Precision-Recall curves of HyCoViT: (a) three-class scenario, (b) four-class scenario.



(a) 3-class scenario



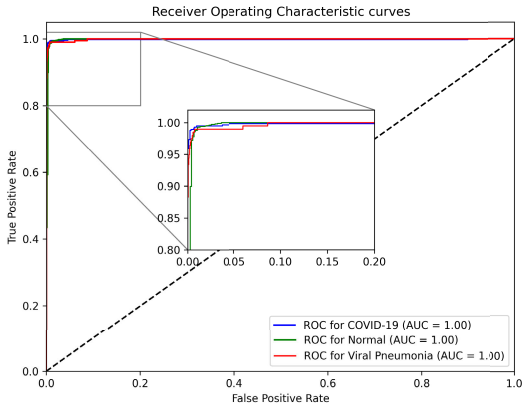
(b) 4-class scenario

FIGURE 6. Confusion Matrix: The horizontal and vertical axes represent the predicted and ground truth labels, respectively. (a) 3-class scenario, (b) 4-class scenario.

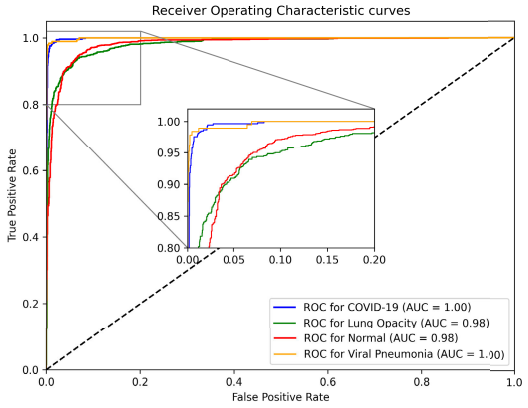
COVID-19 and Lung Opacity, PR curves provide a more informative characterization of classifier behavior, especially in imbalanced settings where ROC curves may be overly optimistic. As shown in Figure 5, we employed a one-vs-all strategy to generate class-wise PR curves and report the corresponding Average Precision (AP) scores, which quantify the trade-off between precision and recall across varying decision thresholds. HyCoViT achieved consistently high AP scores across all classes (e.g., COVID-19: 0.99XX, Pneumonia: 0.99XX in both 3-class and 4-class settings), demonstrating robust detection capability without compromising precision. This is particularly important in medical diagnostics, where high recall is crucial to minimizing false negatives in critical conditions such as COVID-19. The ROC curves presented in Figure 7 further validates these findings for both 3-class and 4-class, with AUC values near to 1.00, confirming excellent class separability. Collectively, these visualizations affirm the strong generalization and clinical utility of HyCoViT for multi-class scenarios.

3) FOUR-CLASS CLASSIFICATION RESULTS PER-CLASS

Table 3 and Figure 7(b) present the per-class results and ROC curve, respectively, for HyCoViT in the four-class classification scenario. Figure 6(b) displays the



(a) Three-class scenario



(b) Four-class scenario

FIGURE 7. ROC curves of HyCoViT: (a) three-class scenario, (b) four-class scenario.

corresponding confusion matrix. As shown in Table 3, HyCoViT achieves a high accuracy over 98% for COVID-19 detection, with a recall rate of 98.58% for COVID-19 and 100% for pneumonia. However, the recall slightly decreases for the Lung Opacity and Normal classes. This indicates that a small proportion of Normal cases, possibly due to COVID-19 or other infections, are incorrectly classified as Lung Opacity, as illustrated in Figure 6(b). Similarly, some Lung Opacity cases are misclassified as Normal. This pattern suggests a notably low number of false-negative cases, which is a highly desirable characteristic in medical data classification. False negatives can lead to the denial or delay of treatment for individuals genuinely affected by a disease. According to Figure 6(b), the overall number of false negatives in COVID-19 detection is very low, with only 13 cases: 5 COVID-19 cases were mistakenly classified as Lung Opacity, and 7 were misclassified as Normal.

The overall accuracy of 96.56% is slightly lower compared to the per-class accuracies. This discrepancy can be attributed to the imbalanced class distribution and marginally higher error rates in the Normal and Lung Opacity classes. However, the strength of our proposed method, particularly in comparison to state-of-the-art deep learning models, are discussed in the following section.

Despite these challenges, HyCoViT demonstrates a strong ability to differentiate between normal and pneumonia cases. Notably, out of 1494 normal cases, 1471 were correctly classified, with none misclassified as pneumonia. However, 4 normal cases were incorrectly classified as COVID-19, and 19 as Lung Opacity. Additionally, the model misclassified only 1 pneumonia case as COVID-19. These results highlight HyCoViT's effectiveness in correctly classifying instances while maintaining a robust balance between precision and recall, leading to a high F1 score.

4) INTERPRETABILITY ANALYSIS

Explainable AI is crucial for medical imaging to enhance transparency and trustworthiness. This study uses COVID-19 detection as a case study to illustrate the effectiveness of XAI techniques, but the findings and methodologies are designed to be generalizable to other medical conditions. Deep Learning models have often been criticized as "black boxes" due to their opaque feature extraction and attention mechanisms, which hinder their credibility in real-world medical applications. To enhance and assess the interpretability of the proposed HyCoViT model, we performed a visualization analysis of attention-weight distributions with and without the inclusion of the custom convolutional block (CB). As shown in Figure 8, the left column presents the original chest X-ray image, the middle column overlays the attention weights extracted from the final encoder layer of HyCoViT with the CB included, and the right column displays the attention map for the same image using HyCoViT without the CB. These attention maps are derived from the final multi-head self-attention layer of the ViT encoder and projected onto the input space to highlight the lung regions that most influenced the model's predictions. Red regions indicate strong attention (task-relevant focus), while blue regions reflect weak or minimal attention. The comparison demonstrates that including the CB leads to more focused attention on clinically significant lung regions, particularly opacities associated with COVID-19, while reducing attention to irrelevant areas. This suggests that the CB enhances spatial localization and contributes to more interpretable decision-making. The improved alignment between attention and pathologically relevant regions supports the diagnostic potential of HyCoViT and underscores its capacity to provide context-aware insights that may aid clinical decision support.

5) COMPARISON OF HYCOVIT WITH SOTA MODELS

To evaluate the performance of the proposed HyCoViT model against state-of-the-art (SOTA) DL models, we conducted a comparative analysis across both three-class and four-class classification scenarios. This comparison includes both CNN-based (C-based) and transformer-based (T-based) models. The results for the three-class classification are presented

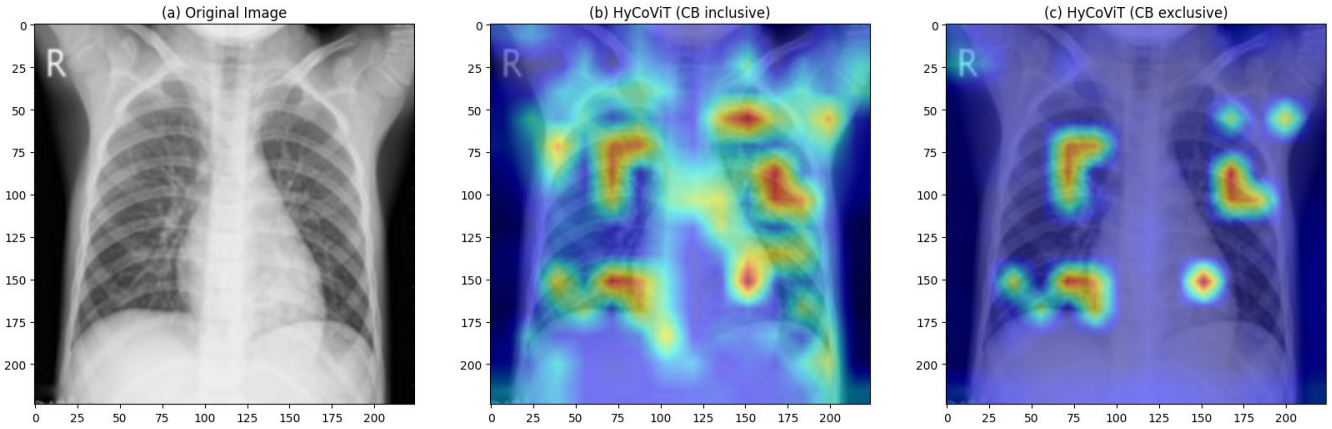
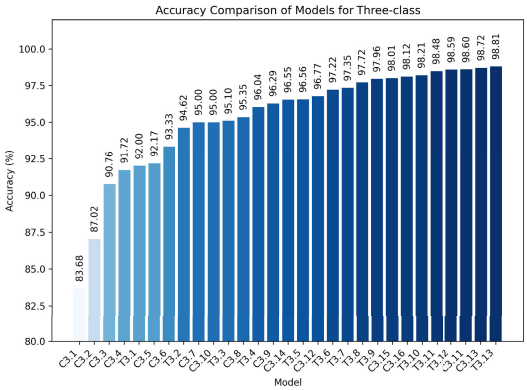


FIGURE 8. Attention-based visualization comparison of a COVID-19 chest X-ray and the corresponding attention maps from the final layer of the HyCoViT model. (a) Left: Original COVID-19 image, (b) Middle: Attention-weight map from HyCoViT with inclusion of the custom convolution block (CB), (c) Right: Attention-weight map from HyCoViT with exclusion of CB. These heatmaps are generated by aggregating the final multi-head self-attention weights from the ViT encoder, highlighting the lung regions most influential in the model's classification decision.

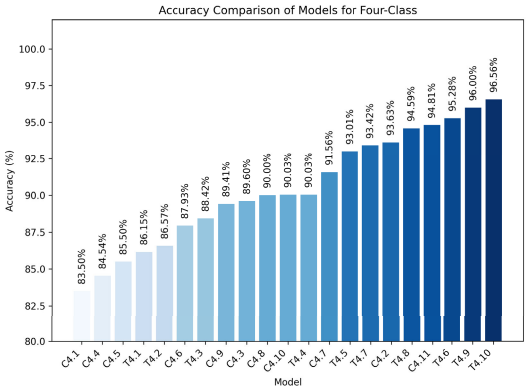
in Table 4, highlighting how HyCoViT performs relative to other SOTA models in these scenarios. Our proposed HyCoViT in the three-class scenario stands out as the leading performer with an Accuracy of 98.81%, Precision of 98.78%, Recall of 98.90%, and an F1-Score of 98.84%. These results demonstrate that HyCoViT not only excels in accurate classification but also achieves a strong balance between Precision and Recall, resulting in a notably high F1-Score. When comparing HyCoViT to the average performance of all C-based models, the improvements are substantial: 4.90% higher in Accuracy, 4.52% higher in Precision, 5.92% higher in Recall, and 5.42% higher in F1-Score. These findings confirm that HyCoViT significantly outperforms C-based SOTA models across all metrics, proving its superiority in both prediction accuracy and the balance between Precision and Recall.

In comparison to the average performance of all T-based models, HyCoViT also demonstrates considerable improvements: 2.05% higher Accuracy, 1.83% higher Precision, 2.23% higher Recall, and 2.19% higher F1-Score. The closest competitor to HyCoViT is IEViT-L/32 [57], which achieved an Accuracy of 98.59% and an F1-Score of 98.05%. Although HyCoViT marginally outperforms IEViT-L/32 in most metrics, it is worth noting that HyCoViT's Precision, at 98.78%, is 0.2% lower than that of IEViT-L/32. This slight difference highlights the overall effectiveness of the HyCoViT architecture. Additionally, the consistently strong performance of T-based models, as shown in Table 4, suggests that these architectures are particularly well-suited for handling complex multi-class classification tasks in medical imaging.

Figure 9 (a) illustrates the accuracy overview for all C-based and T-based models in the three-class scenario. The figure shows that among the top five models with the highest accuracy, T-based models occupy three positions, while C-based models occupy two. This distribution underscores



(a) Three-class scenario



(b) Four-class scenario

FIGURE 9. Accuracy (%) comparison across models, (a) Three-class scenario, (b) Four-class scenario.

the superior performance of T-based models compared to C-based models in these tasks.

Table 5 presents the classification results for the four-class scenario. The results demonstrate that the proposed

TABLE 4. Three-class classification results (%). Bold numbers indicate the best results, and underlined numbers denote the second-best results.

Model		Accuracy	Precision	Recall	F1-score
CNN-Based Models					
C3.1	CNN-block-L [57]	83.68	77.41	72.00	74.24
C3.2	DarkCovidNet [58]	87.02	89.96	85.35	87.37
C3.3	EfficientNet B3-X [59]	90.76	96.11	94.32	95.13
C3.4	DenseNet-201 [60]	91.72	95.24	88.61	92.07
C3.5	Benmalek et al. [61]	92.17	93.45	92.35	92.26
C3.6	Covid-Net [4]	93.33	93.55	93.33	93.36
C3.7	3-Class CoroNet [62]	95.00	95.00	96.90	95.60
C3.8	Fus-ResNet50 [63]	95.35	95.41	95.35	95.34
C3.9	DenseNet201+SVM [64]	96.29	96.42	96.42	96.41
C3.10	EDL-COVID [65]	95.00	95.54	95.27	94.90
C3.11	Al-Sheikh et al. [66]	<u>98.60</u>	98.30	98.40	98.35
C3.12	DeepCOVNet [67]	96.77	97.00	97.00	96.00
C3.13	CovidExpert [68]	98.72	98.72	<u>98.72</u>	<u>98.72</u>
C3.14	Ensemble CNN [69]	96.55	95.55	96.57	96.55
C3.15	Ensemble CNN [70]	98.01	98.77	97.64	98.01
C3.16	ChestCovidNet [71]	98.12	95.75	95.75	95.75
Average of all C-based models:		94.19	94.51	93.37	93.75
Transformer-Based Models					
T3.1	COVID-Transformer [72]	92.00	91.74	95.15	93.40
T3.2	ViT w/ Siamese encoder [73]	94.62	96.77	96.77	96.77
T3.3	MAG-SD [74]	95.10	95.80	94.70	95.20
T3.4	Res50-PA-ViT [75]	96.04	95.70	95.30	95.70
T3.5	Dense121-PA-ViT [75]	96.56	96.00	96.30	96.30
T3.6	DAM-Net [76]	97.22	96.87	98.12	95.54
T3.7	MF-ViT CA [77]	97.35	97.55	97.35	97.40
T3.8	ViT_Base_Eff-B1_224 [78]	97.72	98.00	98.47	98.23
T3.9	ViT-B/16 [77]	97.96	98.77	94.17	96.27
T3.10	CovidViT [79]	98.21	97.85	97.34	97.61
T3.11	Swin Transformer [80]	98.48	98.16	97.91	98.03
T3.12	IEViT-L/32 [57]	98.59	**99.06**	97.09	98.05
T3.13	**HyCoViT (Ours)**	**98.81**	<u>98.78</u>	**98.90**	**98.84**
Average of all T-based models:		96.82	97.00	96.74	96.72

TABLE 5. Four-class classification results (%). Bold numbers indicate the best results, and underlined numbers denote the second-best results.

Model		Accuracy	Precision	Recall	F1-score
CNN-Based Models					
C4.1	Covid-Net [4]	83.50	83.30	87.22	83.92
C4.2	Varalakshmi Perumal et al. [81]	93.63	96.65	93.63	93.63
C4.3	4-Class CoroNet [62]	89.60	90.00	89.92	96.40
C4.4	AlexNet [82]	84.54	87.20	81.70	84.40
C4.5	VGGNet [13]	85.50	89.90	73.20	80.70
C4.6	Wide ReNet [83]	87.93	82.70	82.50	82.60
C4.7	Inception V3 [84]	91.56	83.70	83.30	83.50
C4.8	ConvNeXt [85]	90.00	90.00	81.20	85.40
C4.9	ResNet [86]	89.41	81.20	85.90	83.50
C4.10	CNN-ConvLSTM [17]	90.03	88.70	87.30	88.00
C4.11	NasNetMobile [87]	94.81	95.50	95.00	95.25
Average of all C-based models:		89.14	88.08	85.53	87.03
Transformer-Based Models					
T4.1	Sitaula et al. [21]	86.15	87.48	96.31	86.89
T4.2	Swin Transformer [80]	86.57	85.69	84.14	84.55
T4.3	Swin Transformer w/o Cutout [21]	88.42	88.92	86.38	87.06
T4.4	Pneu Net [23]	90.03	89.58	89.66	89.59
T4.5	ViT-Base-patch16 [20]	93.01	93.48	94.70	94.56
T4.6	CMT (ResNet) [17]	95.28	95.50	92.20	93.80
T4.7	CMT (ConvNeXt) [17]	93.42	93.70	91.60	92.60
T4.8	CMT (Wide ResNet) [17]	94.59	93.90	86.50	90.10
T4.9	xViTCOS-CXR [26]	<u>96.00</u>	<u>96.50</u>	<u>95.90</u>	<u>96.10</u>
T4.10	**HyCoViT (Ours)**	**96.56**	**97.96**	**98.54**	**98.24**
Average of all T-based models:		92.00	92.27	91.59	91.35

HyCoViT model outperforms all C-based and T-based SOTA models in this more complex diagnostic task, which involves distinguishing between nuanced differences among COVID-19, Pneumonia, and Lung Opacity. This highlights HyCoViT's superior capability in handling intricate medical image classifications.

HyCoViT achieves the highest accuracy at 96.56%, which is 8.32% higher than the average accuracy of C-based models (89.14%) and 4.96% higher than the average accuracy of other T-based models (92.00%). Similarly, with a precision of 97.96%, HyCoViT surpasses the C-based average by 11.22% and the T-based average by 6.17%. This performance gap widens even further in the recall metric, where HyCoViT's recall of 98.54% is 15.21% higher than the C-based average (85.53%) and 7.59% higher than the T-based average (91.59%).

The high recall and precision of HyCoViT demonstrate the model's exceptional sensitivity in detecting COVID-19 features while accurately distinguishing other conditions. However, increasing recall can sometimes lead to a decrease in precision, as capturing more true positives might also involve capturing more false positives. Therefore, in this context, the F1-score, which balances recall and precision by taking their harmonic mean, serves as a more reliable metric. HyCoViT's F1-score is 98.24%, which is 12.88% higher than the C-based average (87.03%) and 7.54% higher than the T-based average (91.35%).

In the four-class classification task, T-based models consistently outperform C-based models, as evidenced by their higher average scores across all metrics. Figure 9(b) provides an accuracy overview for all C-based and T-based models in the four-class scenario. The figure shows that among the top five models with the highest accuracy, T-based models occupy four positions, while C-based models occupy only one. This distribution highlights the superiority of T-based models in managing more complex multi-class diagnostic tasks.

C-based models exhibit greater variability in performance, with wider ranges across all metrics, indicating potential instability when handling the more complex four-class classification task. In contrast, T-based models, particularly HyCoViT, demonstrate greater consistency and reliability, as reflected in their narrower performance ranges and higher overall averages. These findings suggest that HyCoViT's architecture is particularly well-suited for distinguishing subtle differences in medical imaging, which is critical in applications where diagnostic accuracy is paramount.

F. ABLATION STUDY

We conducted an ablation study to assess the contribution of various components in our four-class HyCoViT model (T4.10) to the overall accuracy and stability of the model. The key components analyzed in this study were the addition of the convolutional block (CB) and dynamic dropout (DD). The results of this analysis are presented in Table 6.

In addition to these pivotal components, there are numerous subtle and implicit design choices, such as hyperparameters like learning rate schedules and weight decay, that can significantly impact the model's performance. However, these factors are difficult to isolate in a controlled manner. To ensure the reliability and consistency of our findings, we performed the ablation study under a consistent experimental setup, thereby minimizing the influence of uncontrolled variables and focusing on the contributions of the specific components being analyzed.

TABLE 6. Ablation study of the effect of the proposed methods, Convolution Block (CB) and Dynamic Dropout (DD). ✓ denotes that the proposed method is applied.

Model	CB	DD	Acc. (%)
HyCoViT (T4.10)	✗	✗	93.15
HyCoViT (T4.10)	✗	✓	94.21
HyCoViT (T4.10)	✓	✗	95.62
HyCoViT (T4.10)	✓	✓	96.56

TABLE 7. Evaluation of architectural component variants in HyCoViT (T4.10) with CB and DD enabled. Each row isolates the impact of a design choice while keeping other settings fixed.

Normalization	Activation	Loss Function	Acc. (%)
BatchNorm	ReLU	Weighted Cross Entropy	94.45
GroupNorm	ReLU	Weighted Cross Entropy	95.10
GroupNorm	**Mish**	Weighted Cross Entropy	95.73
GroupNorm	Mish	**Focal Loss**	**96.56**

Table 6 summarizes the effects of these components on the accuracy of the HyCoViT model (T4.10). According to the numerical analysis presented in Table 6, the full HyCoViT model achieved the highest accuracy of 96.56%. Removing the CB led to a significant accuracy drop to 94.21%, highlighting its crucial role in the model's architecture for capturing the complex features associated with COVID-19 diagnosis. Similarly, the absence of DD, where the dropout rate was fixed at 0.5, resulted in a decrease in accuracy to 95.62%, underscoring the importance of dropout regulation in preventing overfitting and maintaining model performance.

To further justify the architectural decisions in HyCoViT, we conducted an ablation study focusing on three core components: normalization method, activation function, and loss function. For this analysis, we selected the best-performing configuration, HyCoViT (T4.10), with both the CB and DD enabled. As shown in Table 7, each experiment isolates the effect of a single modification while keeping all other settings fixed. The results clearly demonstrate that using Group Normalization instead of BatchNorm, Mish activation in place of ReLU, and Focal Loss instead of Weighted Cross-Entropy progressively improve performance. These findings validate our selected design choices, which collectively contribute to superior generalization and classification accuracy.

The results presented in Table 8 further validate the strategic placement of DD within the custom CNN block. Applying DD after the second convolutional layer consistently outperformed the alternatives, achieving the highest classification accuracy. This supports the hypothesis that introducing dropout at the mid-level feature stage effectively regularizes the model without disrupting critical early-layer representations or limiting the capacity of later layers.

TABLE 8. Ablation study showing the impact of Dynamic Dropout (DD) placement within the custom CNN block of HyCoViT (T4.10). All models use GroupNorm, Mish activation, and Focal Loss.

DD Placement After	Accuracy (%)
Conv Layer 1	94.18
Conv Layer 2 (ours)	**96.56**
Conv Layer 3	95.63

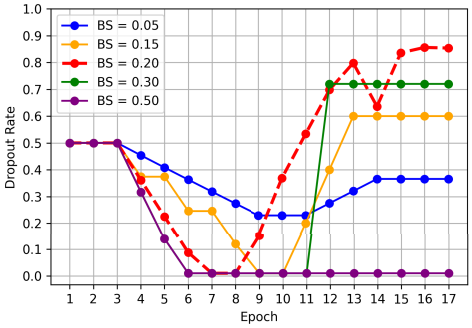


FIGURE 10. Dynamic Dropout (DD) sensitivity analysis for different Base Step (BS) values.

The observed performance trend aligns with the theoretical motivation for balancing feature preservation and adaptive regularization.

DD Sensitivity Analysis over BS Size: In various DL applications, model architectures can range from small to large, each requiring different dropout rates to achieve optimal performance. There is no universal dropout rate that works best across all models. The proposed dynamic dropout rate strategy addresses this by adjusting based on the behavior of training and validation trends. However, the BS parameter in the DD algorithm is critical for balancing learning capacity (model flexibility) and regularization.

In larger pre-trained models, which typically have a vast number of parameters, a larger BS is beneficial as it allows for a quicker reduction in the dropout rate, providing strong regularization to prevent overfitting. This is crucial for complex models that are prone to capturing noise and irrelevant patterns. Conversely, smaller models with fewer parameters benefit from a smaller BS, allowing a more gradual decrease in dropout, which offers the model greater flexibility to learn intricate patterns without being overly constrained.

Figure 10 presents the sensitivity analysis performed on the HyCoViT (T4.10) model with BS as the only variable parameter. The analysis shows that HyCoViT (T4.10) achieved its highest accuracy with a BS value of 0.2, which effectively balances model flexibility at the beginning of training while providing regularization toward the end of the training epoch. A very small value, such as 0.05, did not sufficiently influence the dropout rate to prevent overfitting. On the other hand, a large value, such as 0.5, caused a rapid reduction in the dropout rate per epoch. As illustrated in Figure 10, a BS of 0.5 led to an abrupt reduction in dropout rate within just a few epochs, resulting in overfitting and instability in the HyCoViT (T4.10) model.

V. LIMITATIONS AND FUTURE WORK

The present study demonstrates the effectiveness of HyCoViT on multi-class chest radiograph classification; nevertheless, several avenues remain for extension. Generalizability is the first concern: the model has been evaluated only on CXR data, and its transferability to other imaging modalities such as computed tomography, ultrasound, or digital pathology should be systematically assessed to confirm modality agnostic utility. Second, although our dataset is comparatively large for a curated medical collection, it is still modest relative to natural image benchmarks and originates from a limited number of clinical sites. Future work will therefore focus on assembling multi institutional cohorts that encompass diverse scanners, acquisition protocols, and demographic profiles, thereby enabling a more rigorous evaluation of domain shift and data distribution robustness; advanced augmentation or domain adaptation strategies may also be explored to mitigate inter center variability. Third, investigation on the time and space complexity of the proposed model for resource-constrained clinical settings. Finally, while the attention based heat maps offer intuitive visual explanations, their clinical value must be corroborated through reader studies and prospective trials that measure diagnostic confidence and decision making accuracy among radiologists. Addressing these limitations will be essential for translating HyCoViT from proof of concept to routine clinical deployment.

VI. CONCLUSION

In conclusion, the proposed HyCoViT framework combines the strengths of convolutional and transformer-based architectures to balance local feature extraction with long-range dependency modeling effectively. By introducing a custom convolution block, we address ViT model limitations related to small datasets, enhancing their applicability in medical contexts where data scarcity is common. Additionally, the newly introduced Dynamic Dropout algorithm adaptively adjusts the dropout rate during training, mitigating overfitting and improving model robustness. Our comparative study on three-class and four-class chest X-ray classification tasks, including COVID-19 as a case study, demonstrates that HyCoViT outperforms state-of-the-art CNN-based and transformer-based models. Furthermore, our model successfully identifies opacification areas commonly associated with COVID-19, offering interpretable and clinically meaningful insights into the prediction process. This enhanced interpretability strengthens the diagnostic utility of our system, enabling medical professionals to validate predictions and potentially improve patient outcomes through context-aware decision support. Beyond the results presented, the principles and strategies employed here are broadly applicable to various medical imaging scenarios and can inspire future research. As hybrid architectures continue to evolve, their

capacity to capture both localized and global information, combined with explainability techniques, will likely shape the next generation of interpretable and generalizable solutions in medical image analysis.

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